

✓ Task 5 Report — Applying Security Patches and Updates in Linux

Objective:

The purpose of this task was to ensure that the Linux system remains secure and up to date by applying the latest security patches, software updates, and system upgrades.

Environment Details

- **Operating System:** Kali Linux
 - **System Type:** Virtual Machine (VM)
 - **User Privileges:** Root (sudo)
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Commands Used

Below are the standard commands executed to update and patch the system:

Step 1: Update the package list

```
sudo apt update
```

Step 2: Upgrade all upgradable packages

```
sudo apt upgrade -y
```

Step 3: Perform a full system upgrade including kernel and dependencies

```
sudo apt full-upgrade -y
```

Step 4: Remove any unnecessary packages

```
sudo apt autoremove -y
```

Step 5: (Optional) Clean the package cache

```
sudo apt clean
```

Explanation

Command	Purpose
apt update	Retrieves the latest list of available packages and their versions.
apt upgrade	Installs the newest versions of all packages currently installed.
apt full-upgrade	Handles changing dependencies and installs/removes packages as needed.
apt autoremove	Removes packages that were installed automatically but are no longer needed.
apt clean	Clears out the local repository of downloaded package files.

Verification of System Status

After updates were completed, we verified the current system state using:

Check for held packages or issues

```
sudo apt-mark showhold
```

Check kernel version (if updated)

```
uname -r
```

View last update logs (optional)

```
grep "upgrade" /var/log/dpkg.log
```

Outcome

- All packages were updated to their latest stable versions.
 - No held or broken packages were found.
 - System kernel and core components are up to date.
 - System is secure and ready .
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Security Best Practices Followed

- Updates performed using official Kali Linux repositories.
 - Performed updates in root/sudo context to ensure proper permissions.
 - Verified that no unnecessary services were enabled after the update.
 - Periodic updates are now scheduled as part of system maintenance.
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Conclusion

Keeping the Linux system updated is a foundational step in system security. Through this task, we ensured that our Kali Linux machine is fully patched and less vulnerable to known exploits or system bugs.
